Pathways to Deepseek:
A Peek into the Evolution of LLMs

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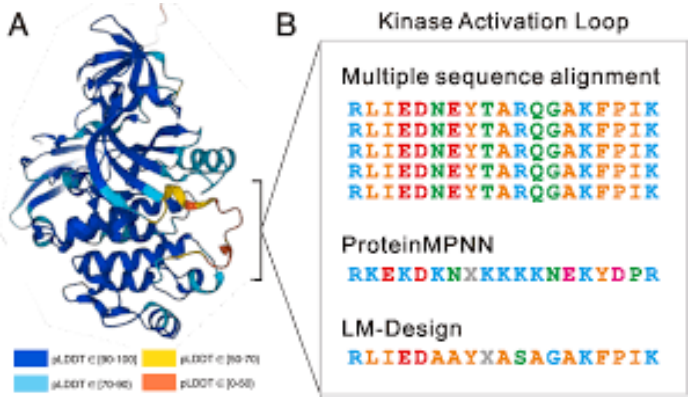
What are large language models (LLMs)

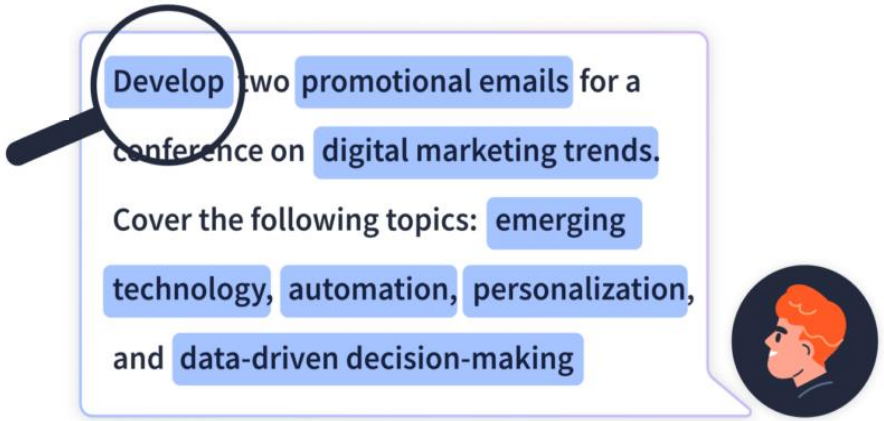
Large language models are AI systems

- they're designed to understand and generate human language.
- they're trained on massive datasets (trillion words), learning patterns, context to produce coherent and meaningful text.

Chatbots and virtual assistants



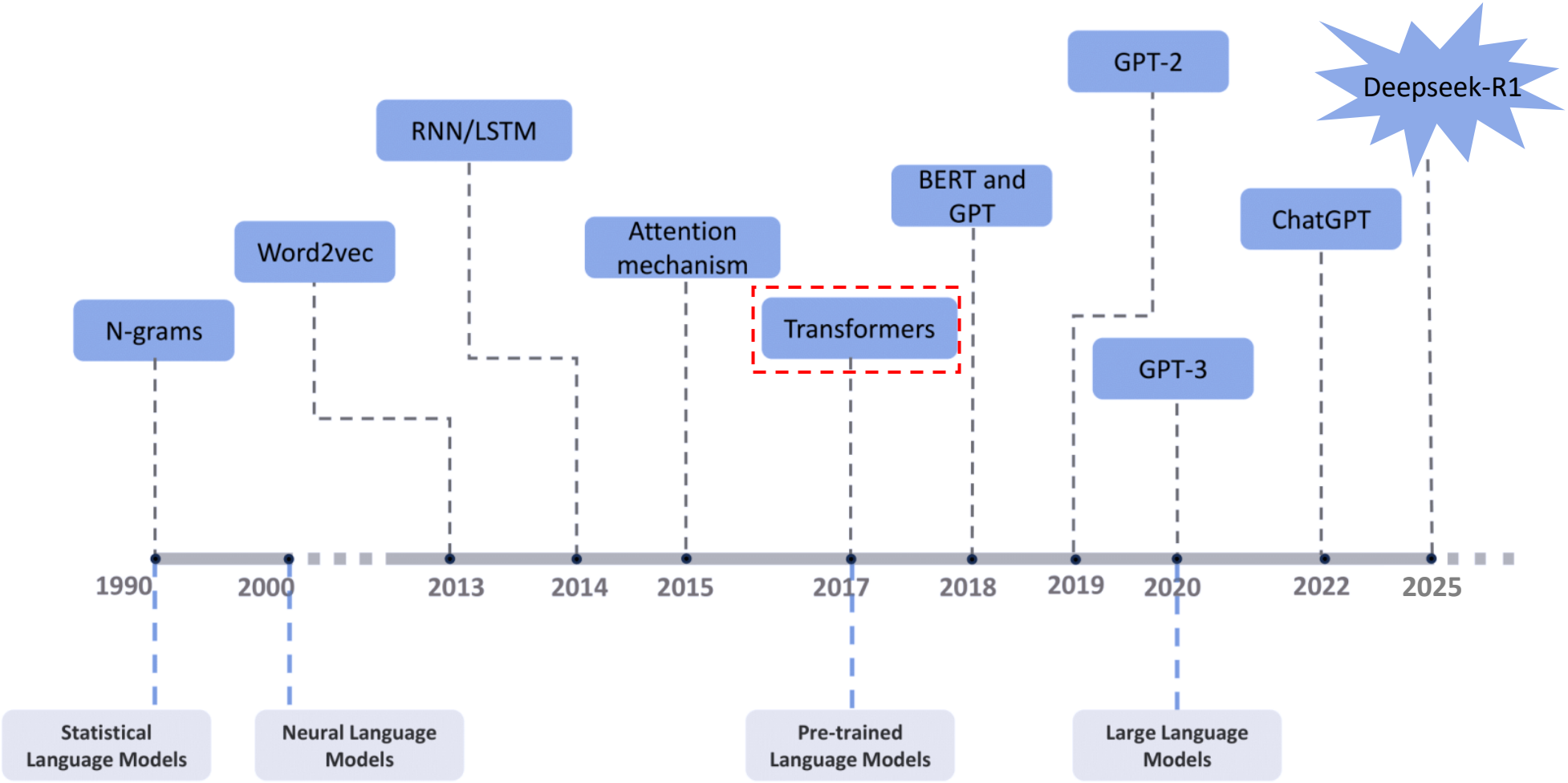
Scientific discovery



Content creation

A brief history of LLMs

From statistical language models (SLM) to large language models (LLM)

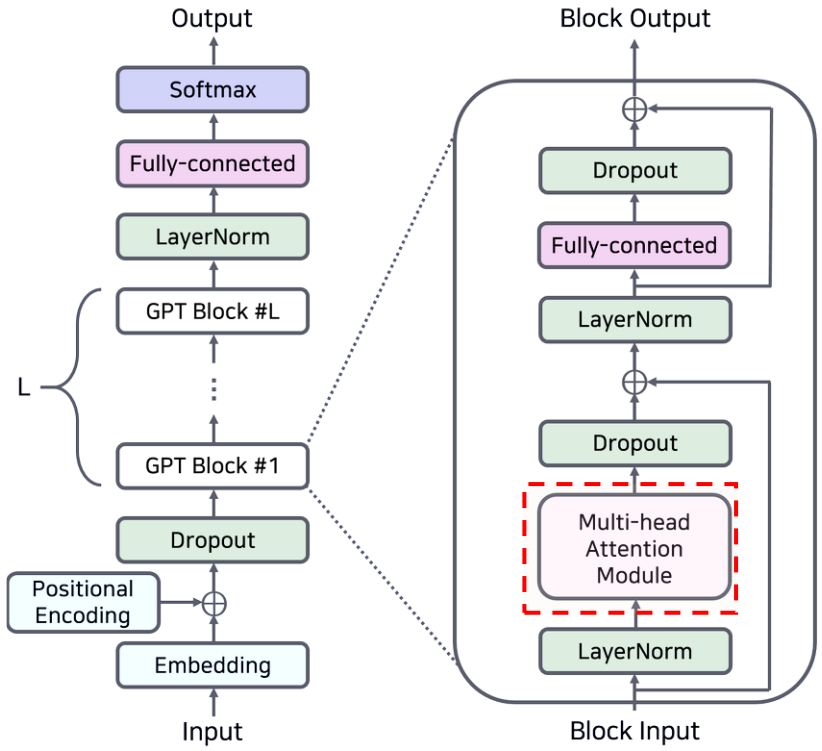


[Chu24] Z. Chu. "History, Development, and Principles of Large Language Models-An Introductory Survey," Arxiv, 2024.

The evolution of the model architecture

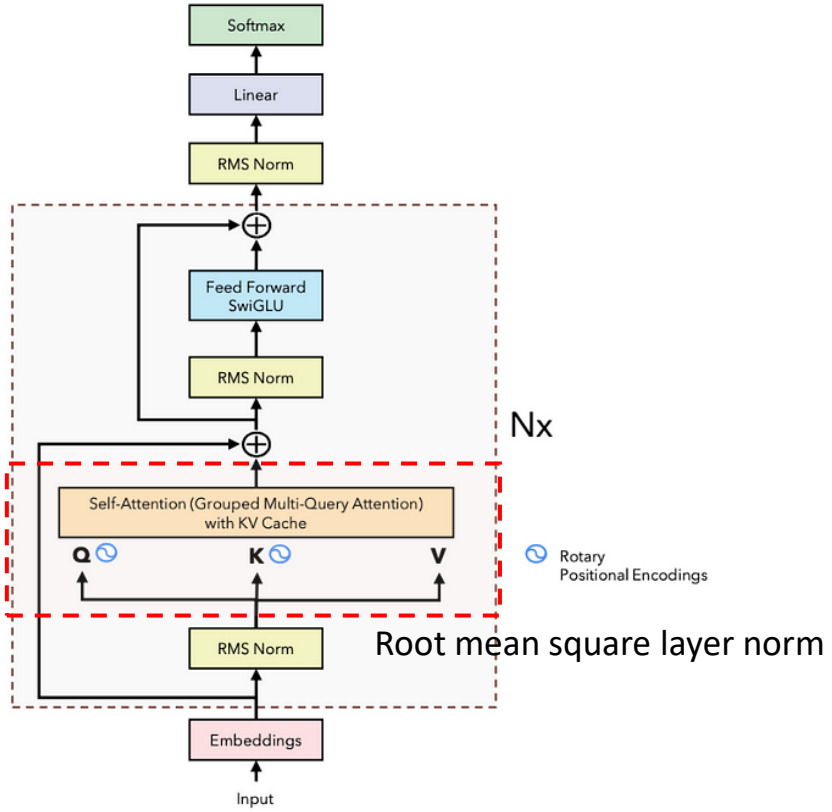
Attention mechanisms (transformers) become the key!

❖ GPT-3 (Generative Pre-training Transformer)



[Brown20] T. Brown, et al. "Language models are few-shot learners.," NeurIPS 2020.

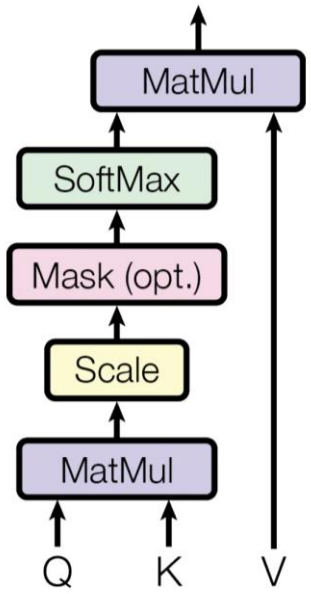
❖ LLaMA-2 (Large Language Model Meta AI)



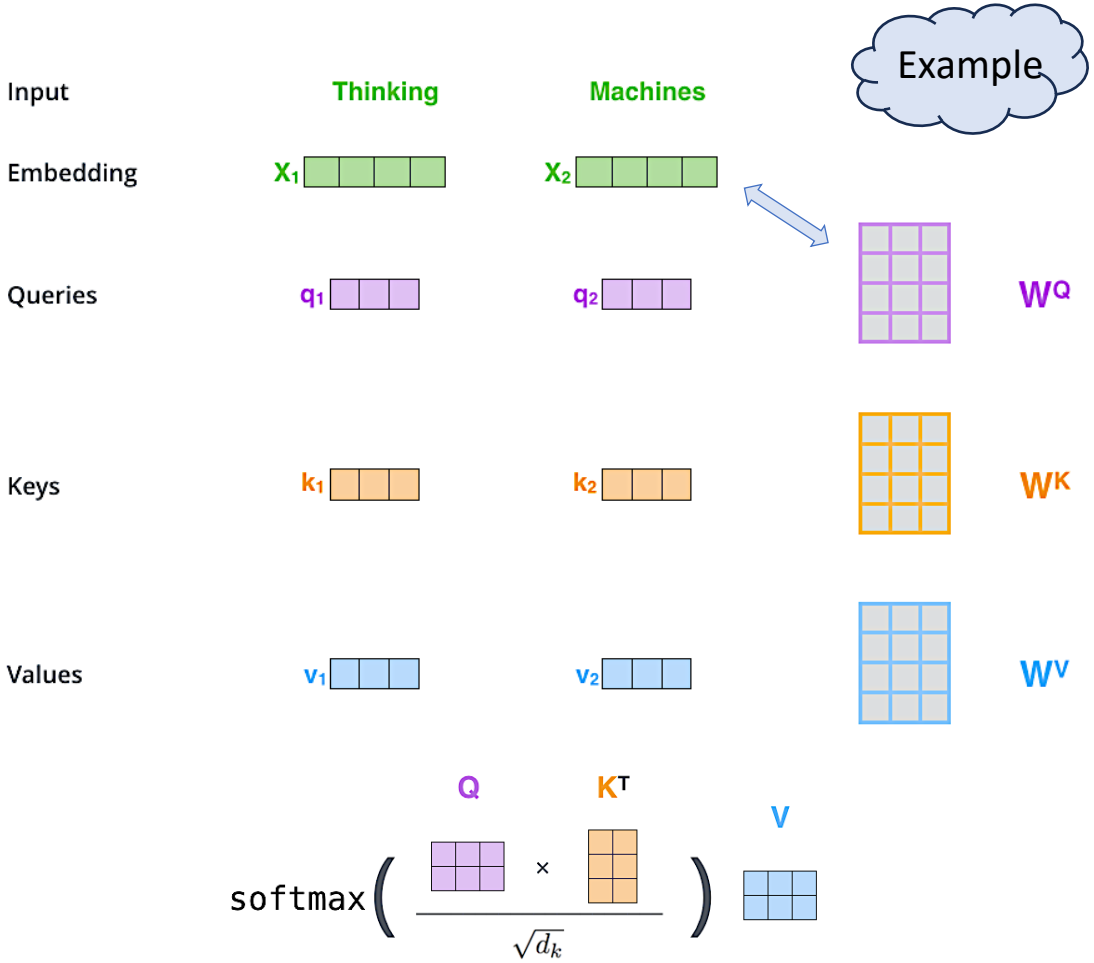
[Brown20] H. Touvron, et al. "Llama 2: Open Foundation and Fine-Tuned Chat Models.," Arxiv, 2023.

Multi-head Attention

The core concept behind transformer: *scaled dot product attention*.



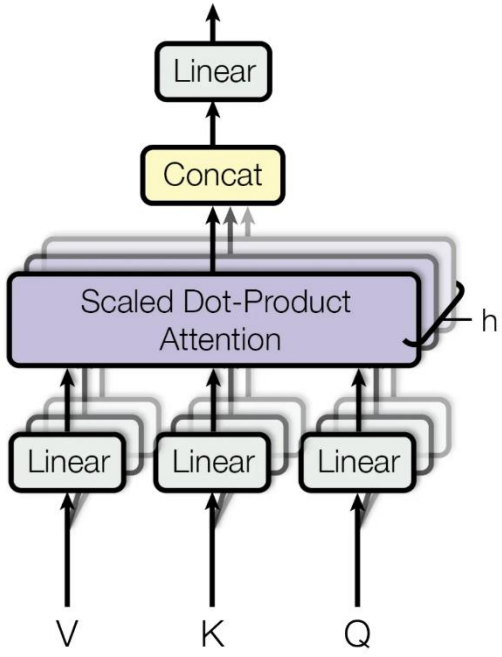
$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$



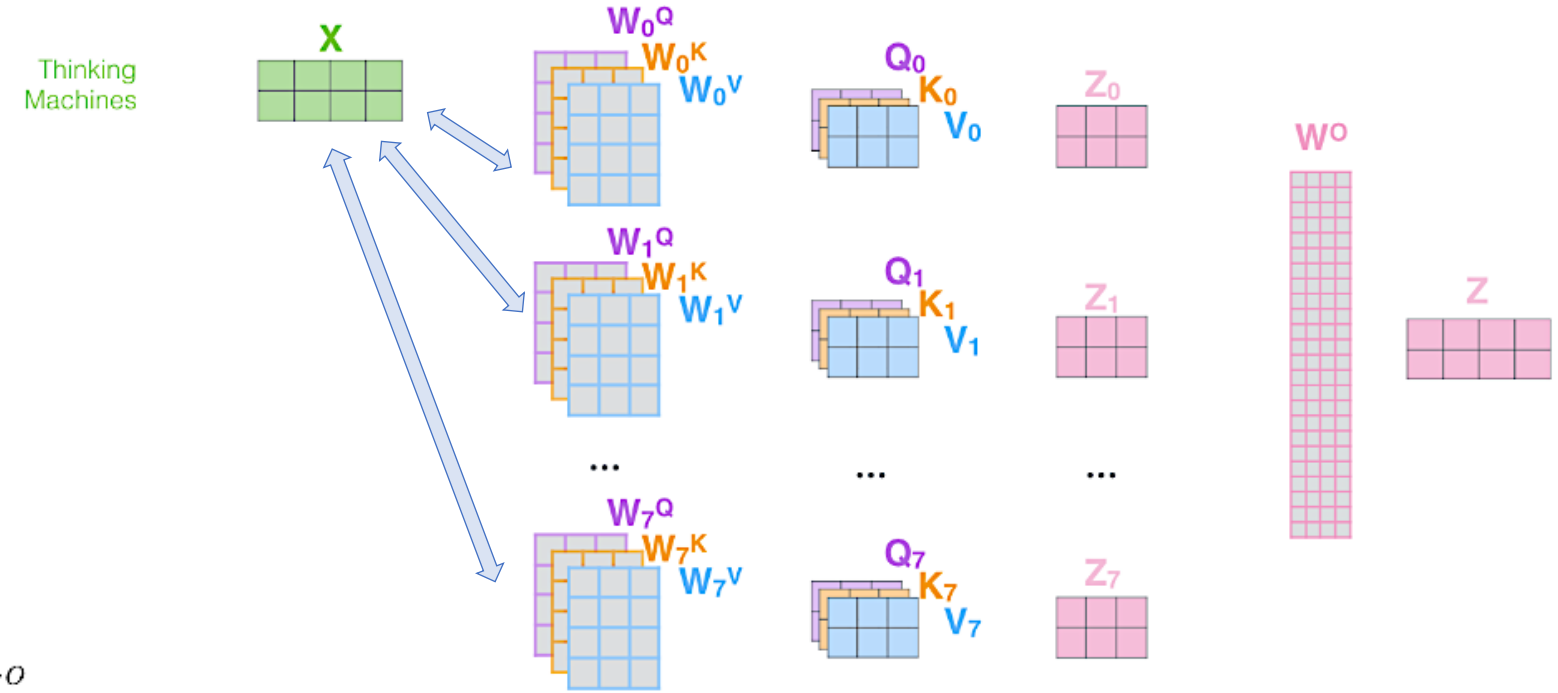
[Vaswani17] A. Vaswani,. “Attention is All you Need,” NeurIPS, 2017.

Multi-head Attention

There are often multiple different aspects a sequence element wants to attend to, and a single weighted average is not a good option for it – This is why we extend the attention mechanisms to multiple heads.

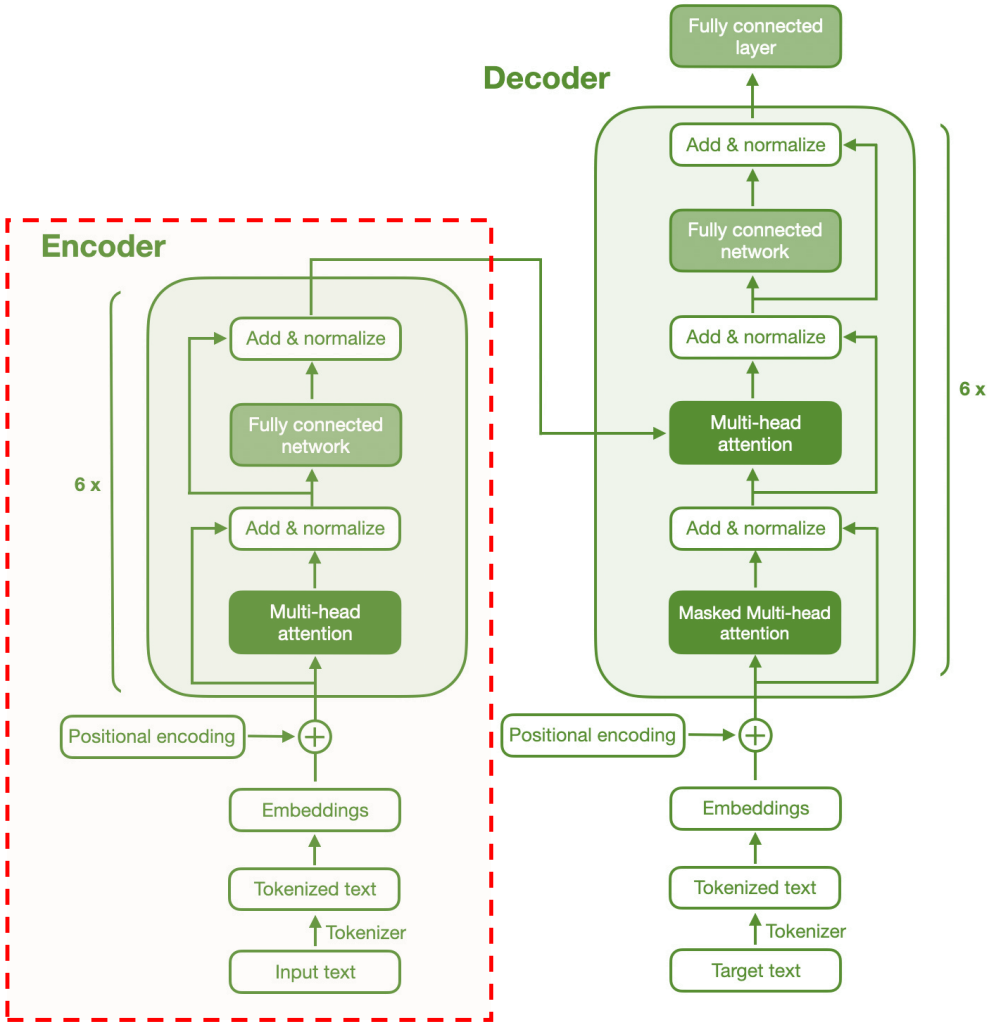


$$\text{Multihead}(Q,K,V) = \text{Concat}(\text{head}_1,\ldots,\text{head}_h)W^O$$
$$\text{where head}_i = \text{Attention}(QW_i^Q, KW_i^K, VW_i^V)$$



[Vaswani17] A. Vaswani,. “Attention is All you Need,” NeurIPS, 2017.

LLMs history: Encoder vs. Decoder



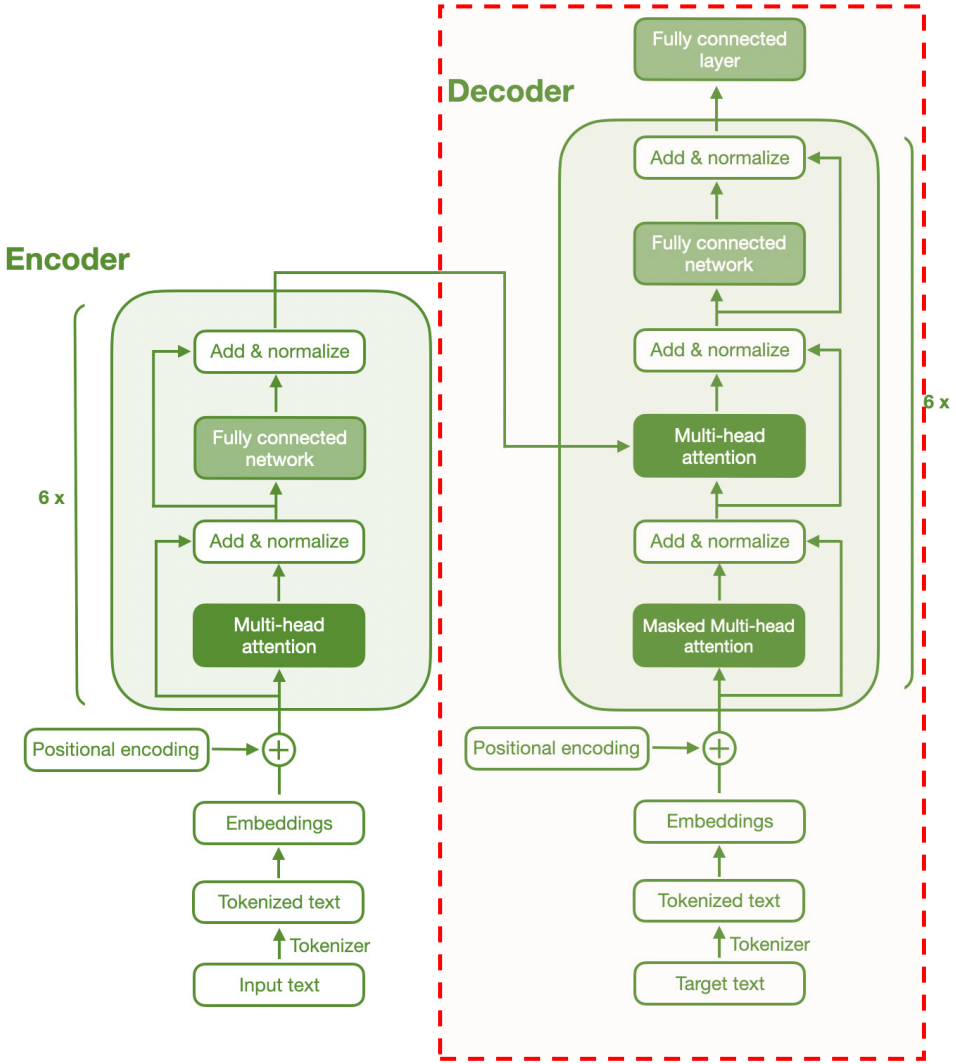
Before 2021, encoder-only transformer models were the mainstream for LLMs.

- Examples: BERT, RoBERTa, ALBERT …
- Training: Next Sentence Prediction & Masked Language Modeling
- They excel in strong contextual understanding
- But less suitable for text generation problems

Type	Model	Size	Provider	Release	Open-sourced	Tuna-bility	Adaptation IT	Adaptation RLHF	Pre-train Data scale
Encoder-only	ALBERT [81]	0.012	Google AI	02/2020	✓	✓	-	-	16GB
	DeBERTa [29]	0.1	Microsoft	06/2020	✓	✓	-	-	80GB
	ELECTRA [86]	0.11	Google AI	03/2020	✓	✓	-	-	-
	BERT [71]	0.11	Google AI	10/2018	✓	✓	-	-	3.3B words
	RoBERTa [87]	0.125	Meta AI	07/2019	✓	✓	-	-	160GB
	XLM-RoBERTa [88]	0.27	Meta AI	07/2020	✓	✓	-	-	2.5TB

[Chu24] Z. Chu. “History, Development, and Principles of Large Language Models-An Introductory Survey,” Arxiv, 2024.

LLMs history: Encoder vs. Decoder



After 2021, decoder-only transformer models dominated the architectures of LLMs.

- ❖ Examples: GPT series, Llama 3.1, Qwen
- ❖ Training: Next Word Prediction
- ❖ These models are known for their excellent text-generation capabilities

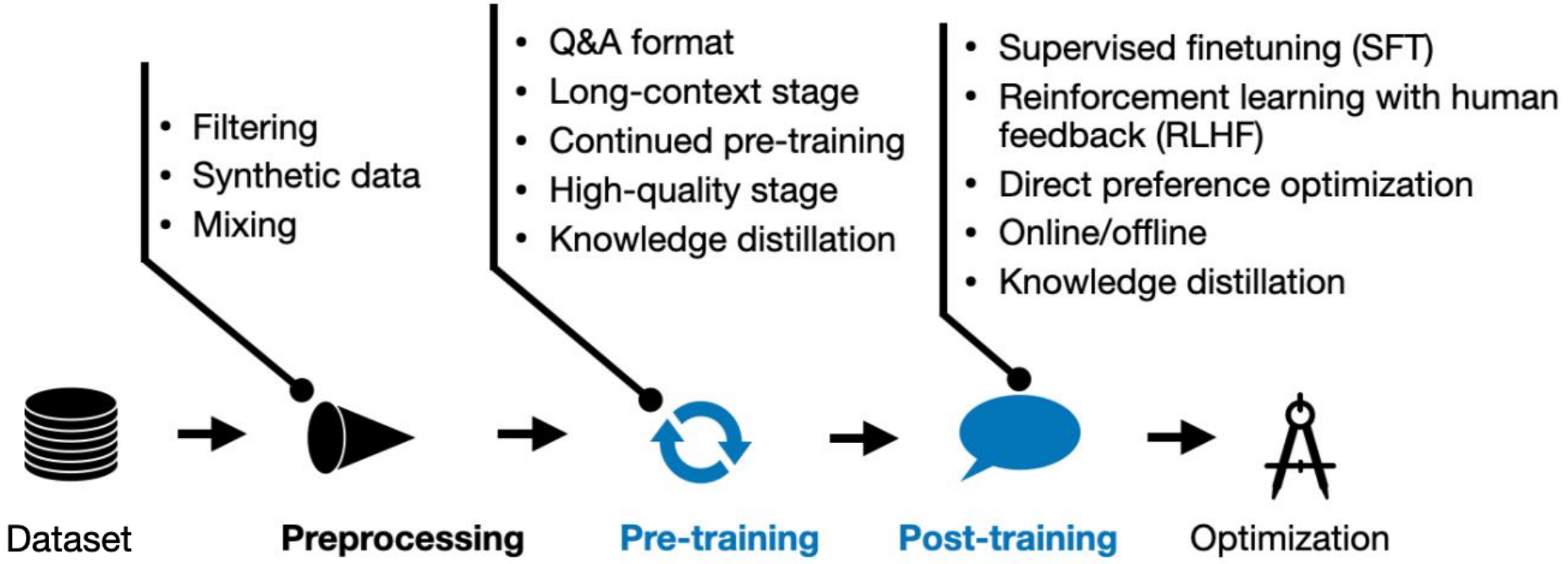
Type	Model	Size	Provider	Release	Open-sourced	Tuna-bility	Adaptation IT	Adaptation RLHF	Pre-train Data scale
Decoder-only	Codex [85]	12	OpenAI	07/2021	x	x	-	-	100B tokens
	Pythia [32]	12	Eleuther AI	04/2023	✓	x	-	-	300B tokens
	CodeGeeX [98]	13	Tsinghua U.	09/2022	✓	✓	-	-	850B tokens
	Skywork [99]	13	Kunlun Inc	10/2023	✓	✓	-	-	3.2T tokens
	QWEN [100]	14	Alibaba	09/2023	✓	✓	✓	✓	3T tokens
	CodeGen2 [101]	16	Salesforce AI	05/2023	✓	✓	-	-	400B tokens
	GPT-NeoX-20B [102]	20	EleutherAI	04/2022	✓	x	-	-	825GB
	Gemini 1.5 [103]	20	Google AI	02/2024	x	✓	-	-	-
	LlaMa 2 [72]	70	Meta AI	07/2023	✓	✓	✓	✓	2T tokens
	LlaMa 3 [104]	70	Meta AI	04/2024	✓	✓	✓	✓	15T tokens
	HyperCLOVA [17]	82	Naiver	09/2021	x	✓	-	-	300B tokens
	Galactia [105]	120	Meta AI	11/2022	✓	✓	-	-	106B tokens
	GLM [28]	130	Tsinghua U.	10/2022	✓	✓	-	-	400B tokens
	LaMDA [106]	137	Google AI	01/2022	x	✓	-	-	768B tokens
	Jurassic-1 [107]	178	AI21 Labs	08/2021	x	✓	-	-	300B tokens
	GPT-3 [10]	175	OpenAI	05/2020	x	x	-	-	300B tokens
	WebGPT [108]	175	OpenAI	12/2021	x	x	-	-	-
	InstructGPT [109]	175	OpenAI	03/2022	x	x	✓	✓	-

……

[Chu24] Z. Chu. "History, Development, and Principles of Large Language Models-An Introductory Survey," Arxiv, 2024.

LLM training paradigms

- **Preprocessing** (Enhances data quality and model efficiency)
- **Pre-training** (Builds foundational knowledge and language understanding)
- **Post-training** (Refines outputs and improves alignment with human intent)

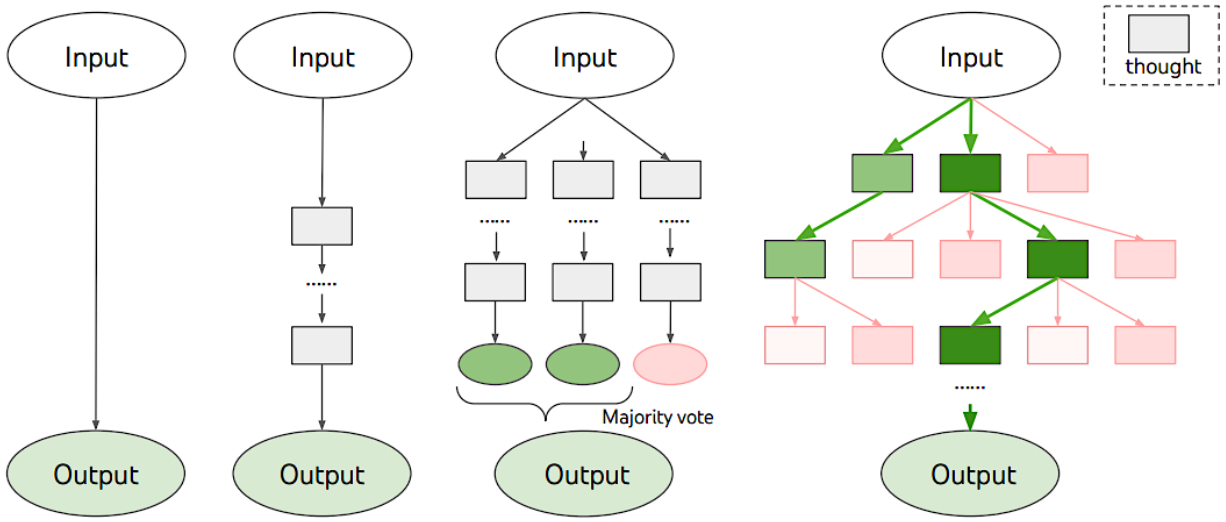


[pulse1] https://www.linkedin.com/pulse/new-llm-pre-training-post-training-paradigms-sebastian-raschka-phd-l53zc

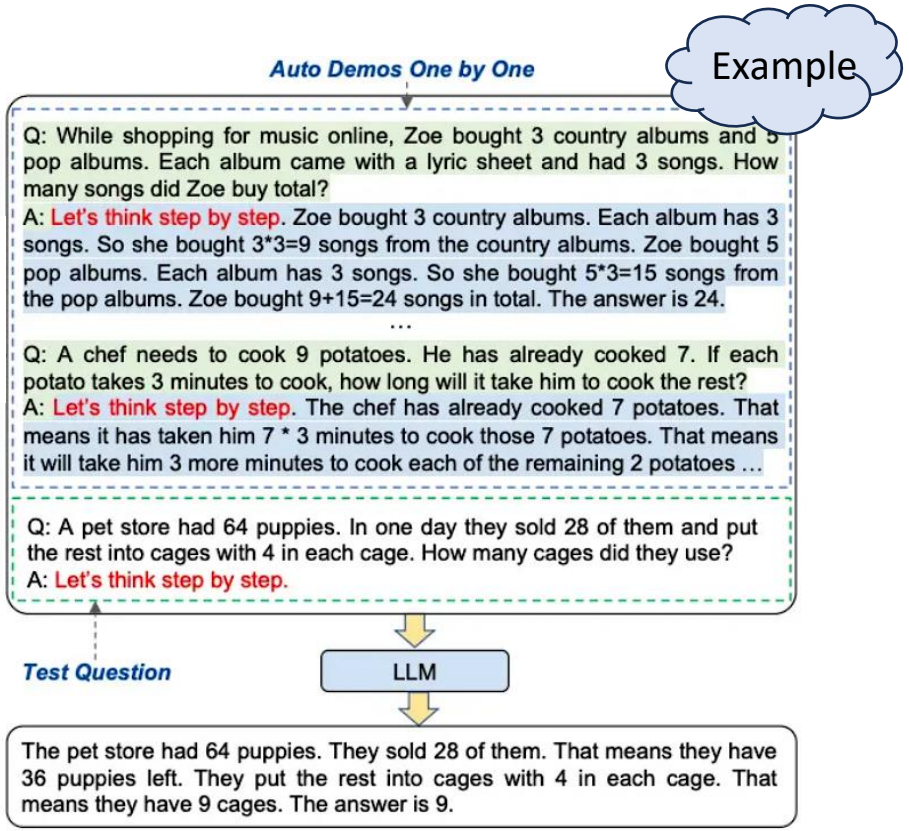
LLM prompting: Chain-of-thoughts (CoT)

Chain-of-Thought (CoT) prompting encourages LLMs to break down a complex "thought" into intermediate steps with a few demonstrations.

❖ **CoT variants**



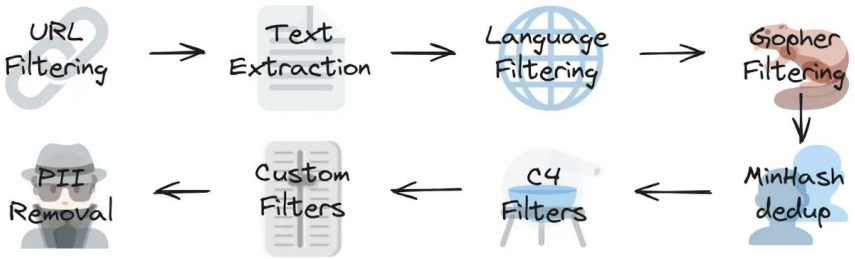
(a) Input-Output Prompting (IO) (c) Chain of Thought Prompting (CoT) (c) Self Consistency with CoT (CoT-SC) **(d) Tree of Thoughts (ToT)**



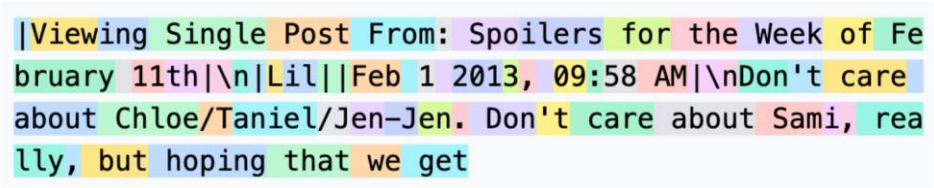
[Zhang22] Z. Zhang, "Automatic Chain of Thought Prompting in Large Language Models", Arxiv, 2022.

LLM Pre-training

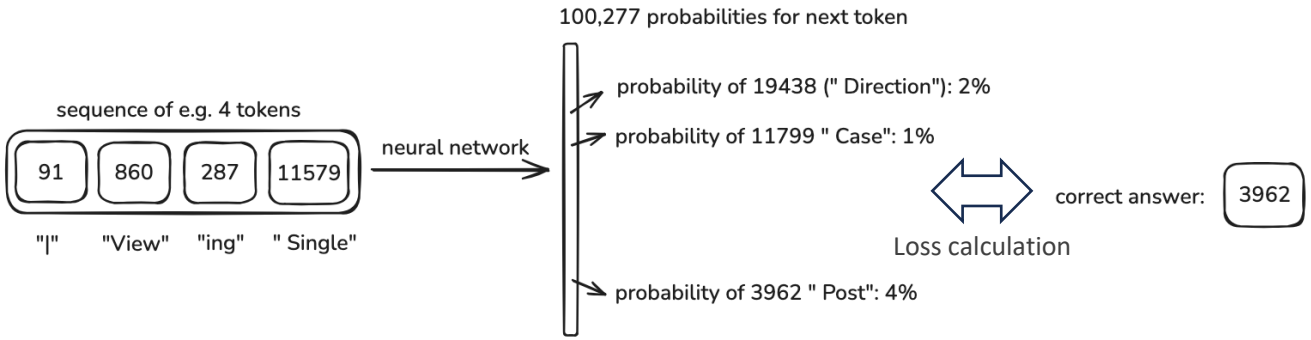
1. **Data Curation**: collected data from the internet (trillion-level words):



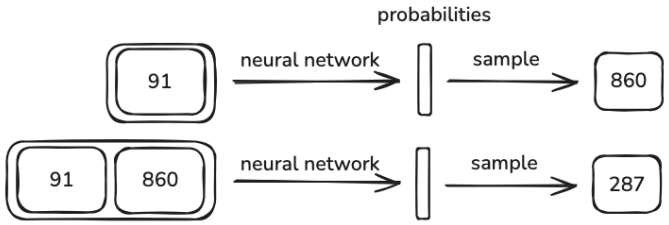
2. **Tokenization**: converting large sequences of text into smaller, manageable units:



3. **Network Training**: probability distribution prediction



4. **Inference**: generating tokens step by step using learned knowledge, following statistical patterns from training data:



LLM Post-training

Supervised Fine-tuning (SFT): uses labeled data to train LLMs for specific improvements, e.g., chatting, counting, reasoning, hallucinating, etc.

Hallucinations

train time

Human: "Who is Tom Cruise?"
Assistant: "Tom Cruise is a famous American actor and producer..."

Human: "Who is John Barrasso?"
Assistant: "John Barrasso is American physician and politician..."

Human: "Who is Genghis Khan?"
Assistant: "Genghis Khan was the founder of the Mongol Empire."

...

test time

Human: "Who is Orson Kovacs?"
Assistant: jdawdawdawd

Bunch of other small random stuff

Which is bigger 9.11 or 9.9?

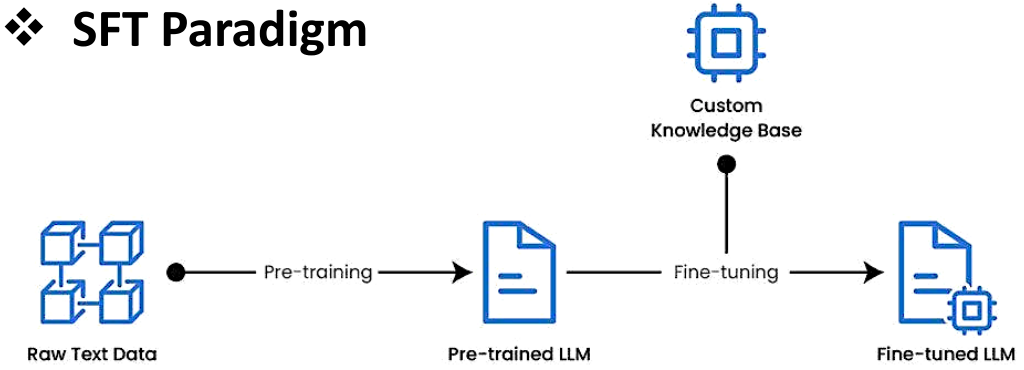
Models can (and should!) use tools!

Web search
Code (/ Python interpreter)

Conversations

Human: "What is 2+2?"
Assistant: "2+2 = 4"
Human: "What if it was * instead of +?"
Assistant: "2*2 = 2"

❖ **SFT Paradigm**



[Karpathy1] A. Karpathy. https://www.youtube.com/watch?v=7xTGNNLPyMI

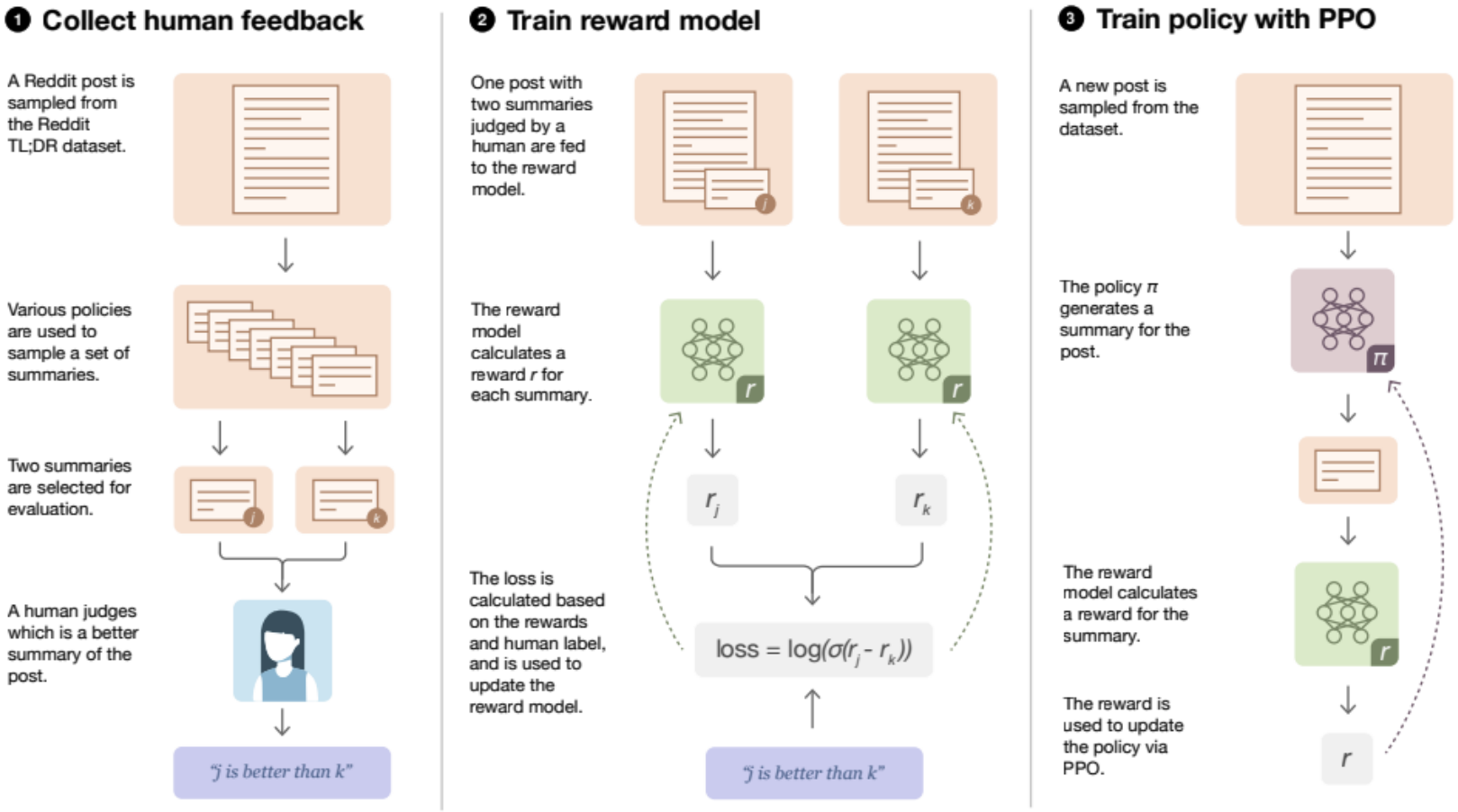
LLM Post-training

How about unverifiable problems? **Reinforcement learning with human feedback** (RLHF)

1. Collect **human feedback** on unverifiable problems, e.g., “write a joke”, and “write a poem”.

2. Train **rewarder** based on preference data from alternative outputs from the LLM.

3. Reward model works to update an LM policy that guides the **reinforcement learning** of LLM.

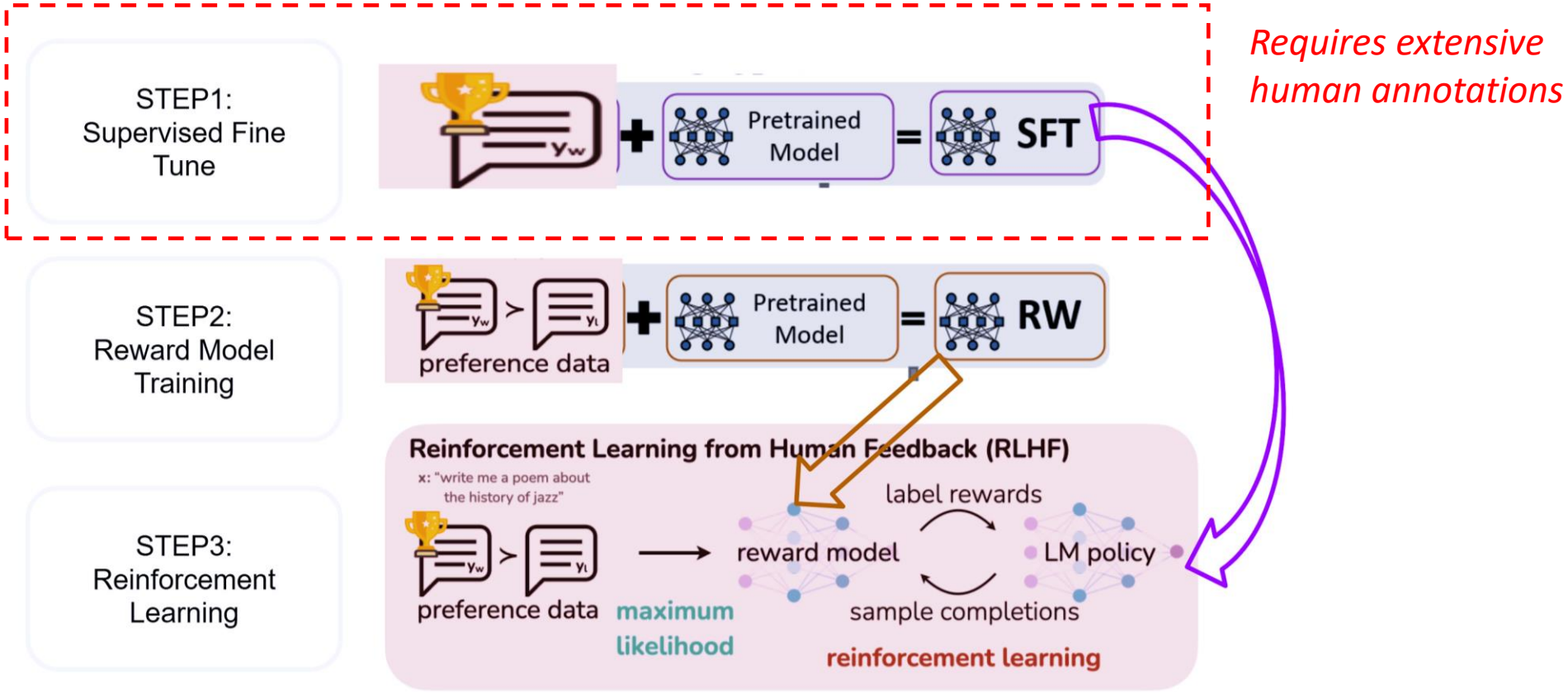


[Ouyang1] L. Ouyang, “Training language models to follow instructions with human feedback. ” NeurIPS 2022

The problem LLM post-training

"Skipping or cutting down on human feedback—that's a big thing," – I. Friedman

LLMs heavily relies on **supervised fine-tuning** (SFT) during post-training phase.



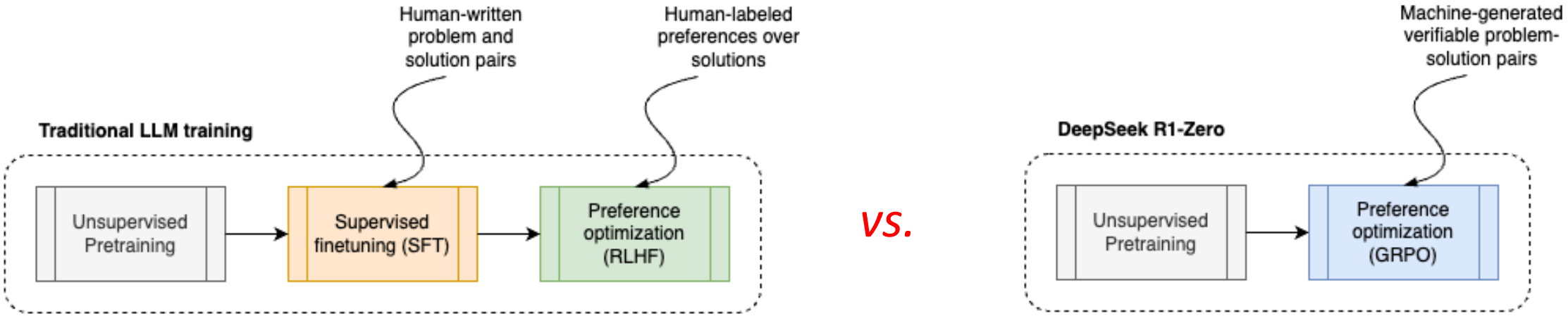
Requires extensive human annotations

[Ouyang1] L. Ouyang, "Training language models to follow instructions with human feedback. " NeurIPS 2022

The revolution of DeepSeek

Breakthrough of Deepseek models

- **DeepSeek-R1-Zero** was trained solely via large-scale RL, skipping supervised fine-tuning (SFT).



Which one is better?

[Deepseek1] Deepseek-AI. “DeepSeek-R1: Incentivizing Reasoning Capability in LLMs via Reinforcement Learning”, Arxiv, 2025.

The revolution of DeepSeek

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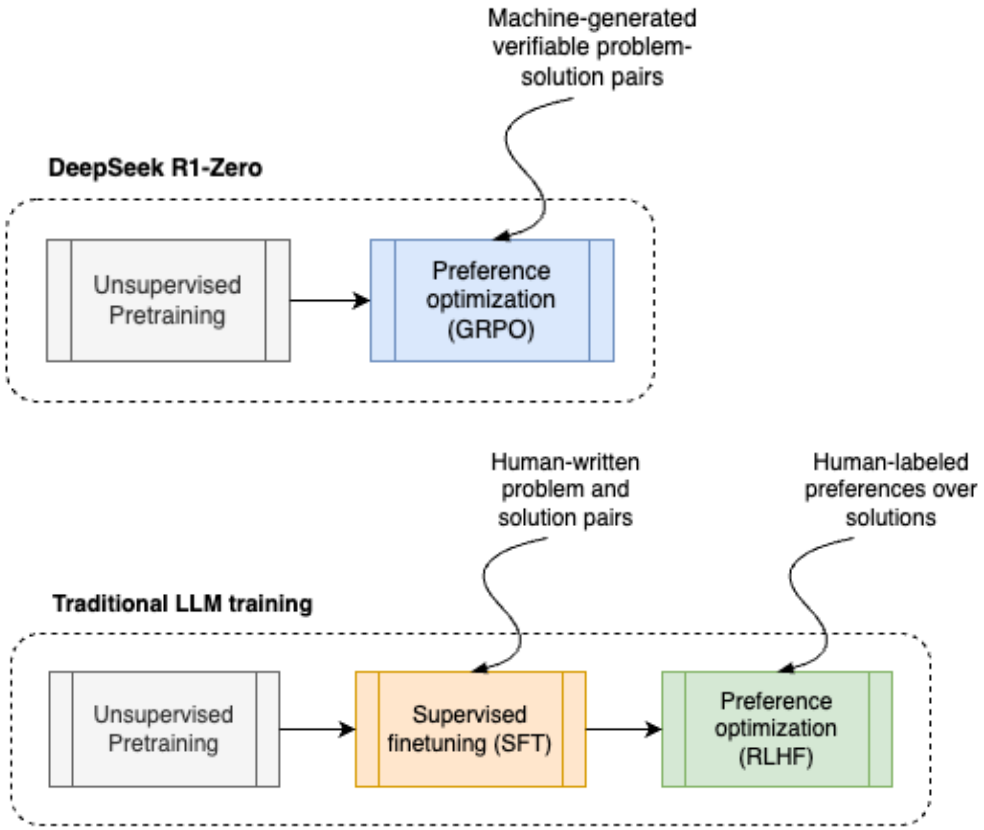
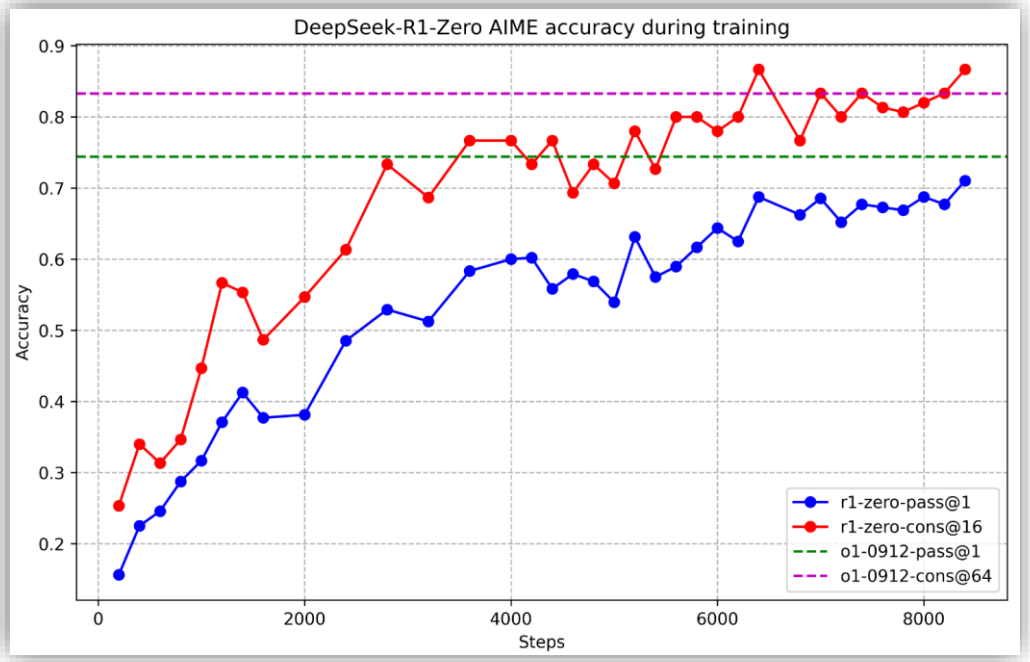


Deepseek r1-zero outperforms OpenAI o1



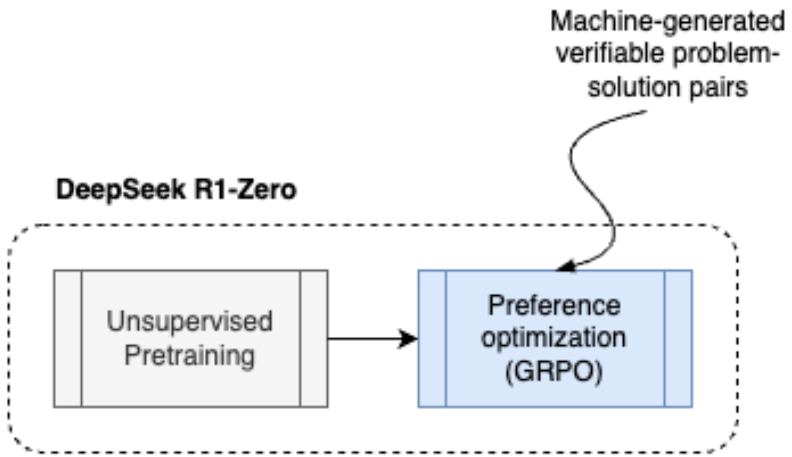
[Deepseek1] Deepseek-AI. “DeepSeek-R1: Incentivizing Reasoning Capability in LLMs via Reinforcement Learning”, Arxiv, 2025.

The revolution of DeepSeek

Breakthrough of Deepseek models

❖ **DeepSeek-R1-Zero** was trained solely via large-scale RL, skipping supervised fine-tuning (SFT).

Yet, **purely RL** also causes several problems:



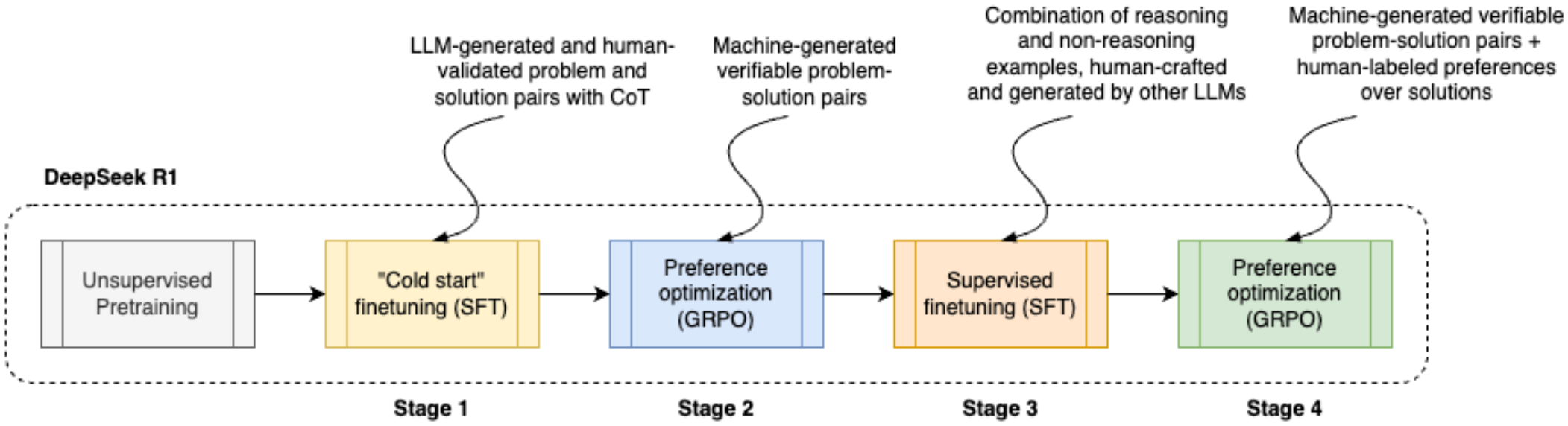
- **Poor readability**—R1-Zero's outputs were often difficult to interpret despite being logically sound.
  1. Inconsistent formatting and structure in responses
  2. Difficulty in following the reasoning chain
- **Language mixing**—R1-Zero would frequently:
  1. Switch between languages (particularly English and Chinese) within a single response
  2. Show inconsistent language use across different parts of the reasoning process

[Deepseek1] Deepseek-AI. “DeepSeek-R1: Incentivizing Reasoning Capability in LLMs via Reinforcement Learning”, Arxiv, 2025.

The revolution of DeepSeek

Breakthrough of Deepseek models

- **DeepSeek-R1-Zero** was trained solely via large-scale RL, skipping supervised fine-tuning (SFT).
- **DeepSeek R1** incorporates a novel four-stage training pipeline designed to maintain strong reasoning capabilities while addressing the limitations of pure RL.



[Deepseek1] Deepseek-AI. “DeepSeek-R1: Incentivizing Reasoning Capability in LLMs via Reinforcement Learning”, Arxiv, 2025.

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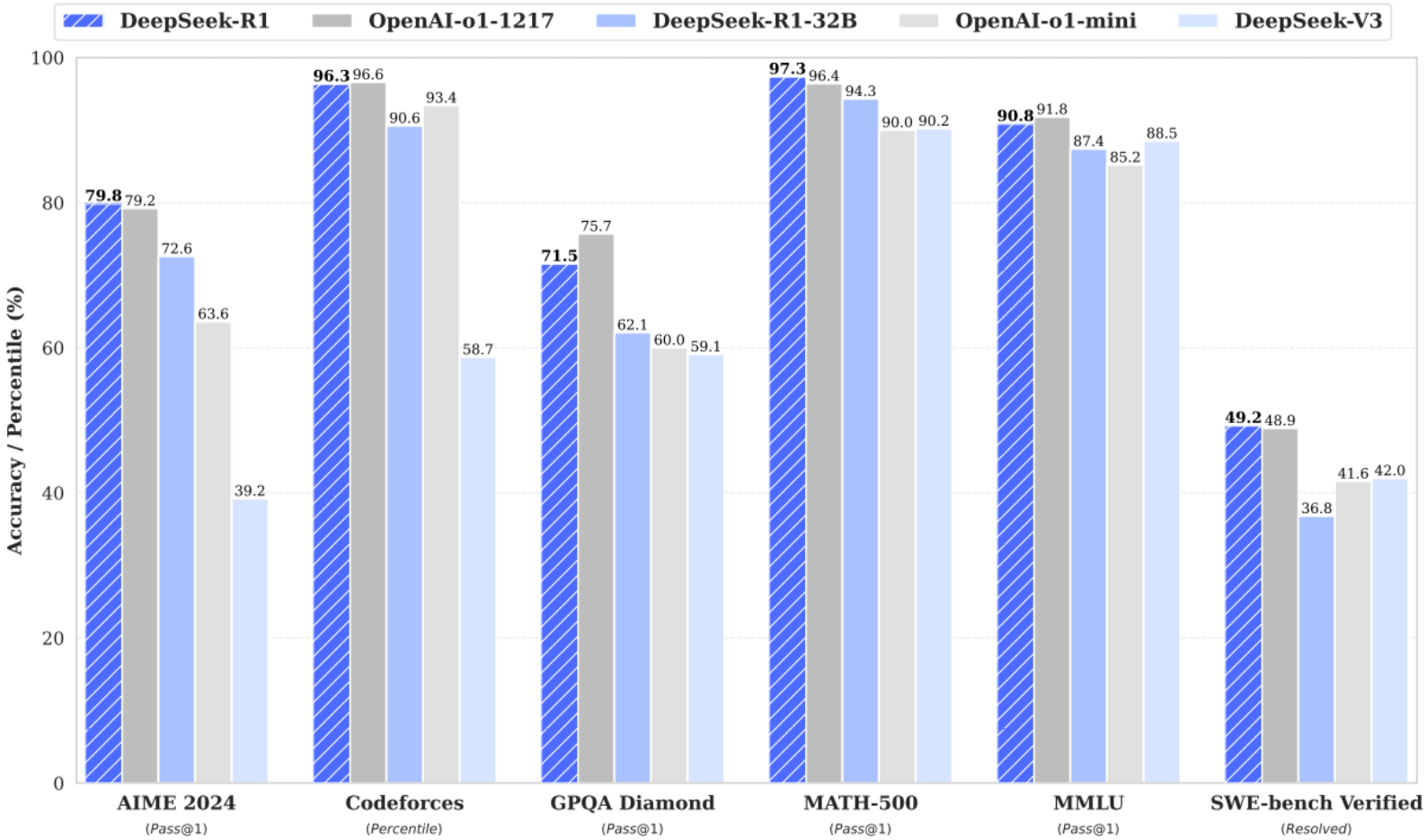
Deepseek-R1 vs. OpenAI-o1



[Deepseek1] Deepseek-AI. “DeepSeek-R1: Incentivizing Reasoning Capability in LLMs via Reinforcement Learning”, Arxiv, 2025.

The revolution of DeepSeek

香港城市大學
City University of Hong Kong

The Distillation Surprise of DeepSeek R1

- **Distillation:** Fine-tuning on 800K samples generated by DeepSeek-R1
- **14B model** outperformed QwQ-32B-Preview
- **1.5B model** hit 28.9% on AIME, beating GPT-4o on certain tasks

Model	AIME 2024		MATH-500	GPQA Diamond	LiveCode Bench	CodeForces
	pass@1	cons@64	pass@1	pass@1	pass@1	rating
GPT-4o-0513	9.3	13.4	74.6	49.9	32.9	759
Claude-3.5-Sonnet-1022	16.0	26.7	78.3	65.0	38.9	717
OpenAI-o1-mini	63.6	80.0	90.0	60.0	53.8	**1820**
QwQ-32B-Preview	50.0	60.0	90.6	54.5	41.9	1316
DeepSeek-R1-Distill-Qwen-1.5B	28.9	52.7	83.9	33.8	16.9	954
DeepSeek-R1-Distill-Qwen-7B	55.5	83.3	92.8	49.1	37.6	1189
DeepSeek-R1-Distill-Qwen-14B	69.7	80.0	93.9	59.1	53.1	1481
DeepSeek-R1-Distill-Qwen-32B	**72.6**	83.3	94.3	62.1	57.2	1691
DeepSeek-R1-Distill-Llama-8B	50.4	80.0	89.1	49.0	39.6	1205
DeepSeek-R1-Distill-Llama-70B	70.0	**86.7**	**94.5**	**65.2**	**57.5**	1633

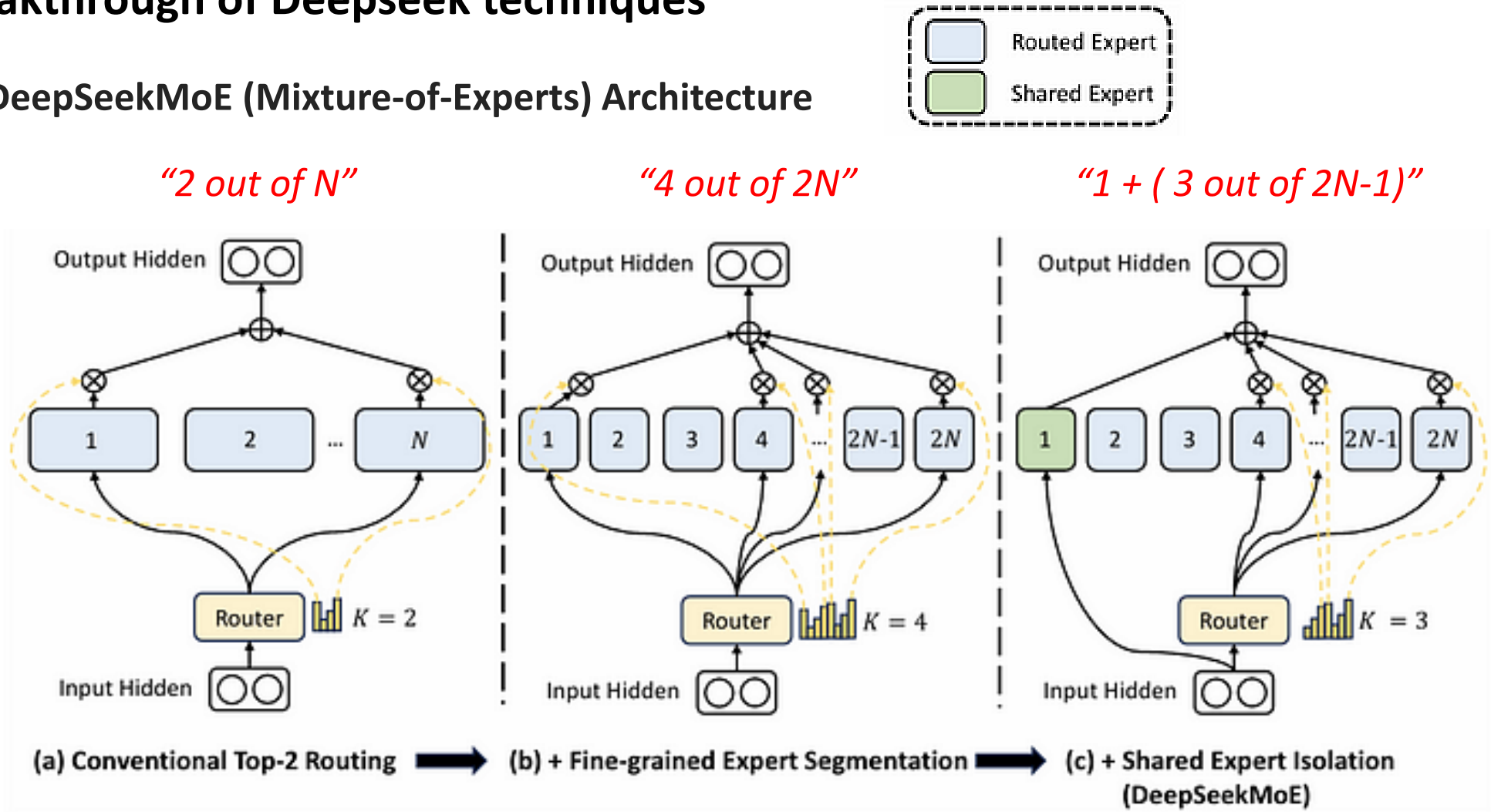
Table 5 | Comparison of DeepSeek-R1 distilled models and other comparable models on reasoning-related benchmarks.

[Deepseek1] Deepseek-AI. “DeepSeek-R1: Incentivizing Reasoning Capability in LLMs via Reinforcement Learning”, Arxiv, 2025.

The revolution of DeepSeek

Breakthrough of Deepseek techniques

❖ **DeepSeekMoE (Mixture-of-Experts) Architecture**



[Deepseek2] Deepseek-AI. "DeepSeek-V3 Technical Report", Arxiv, 2025.

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Breakthrough of Deepseek techniques

❖ **Group Relative Policy Optimization (GRPO)**

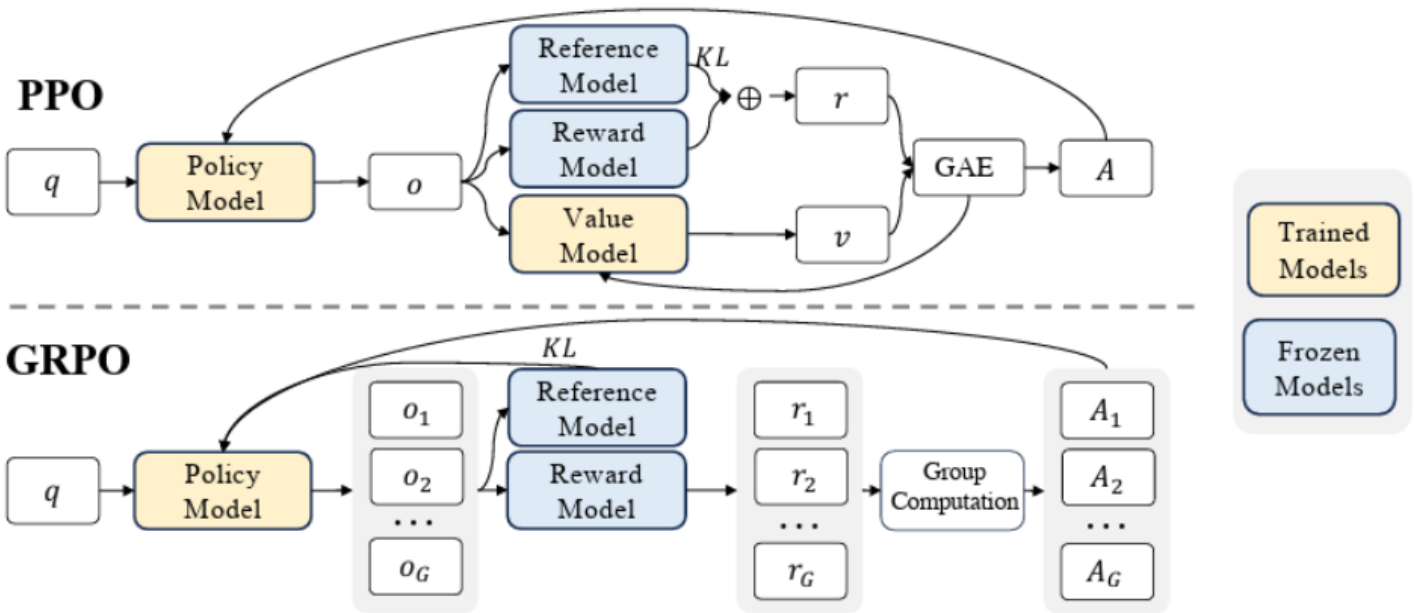


Figure 4 | Demonstration of PPO and our GRPO. GRPO foregoes the value model, instead estimating the baseline from group scores, significantly reducing training resources.

- GRPO is a variant of PPO that **eliminates the need for a value model** while maintaining effective RL.
- GRPO samples multiple outputs for each input and uses the average reward within each group as the baseline.
- GRPO provides a natural way, allowing the model to learn more effectively from its own outputs.

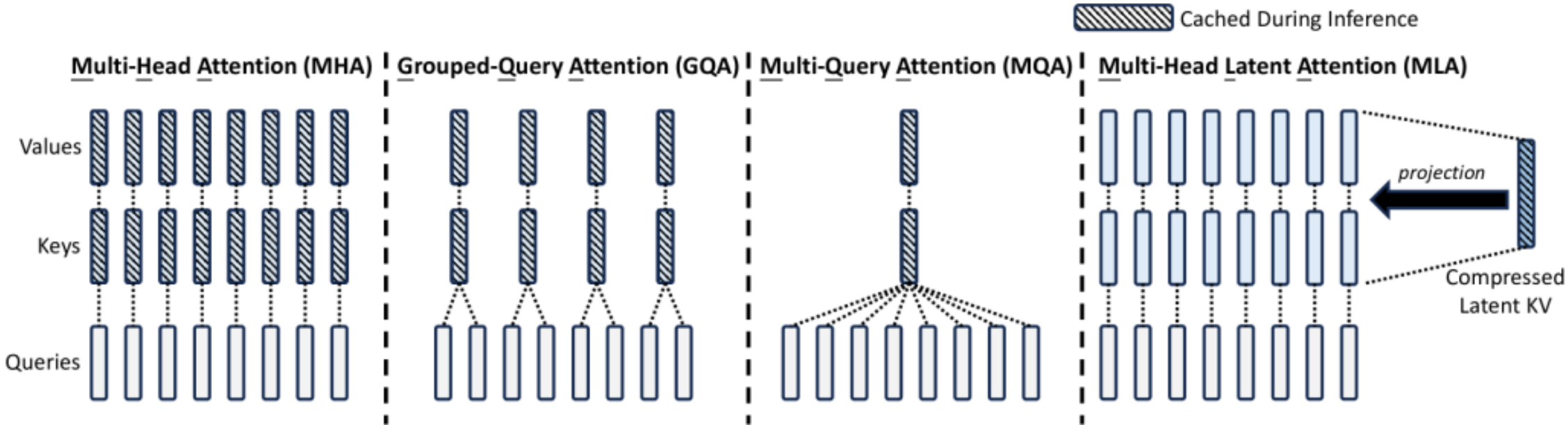
[Deepseek2] Deepseek-AI. “DeepSeek-V3 Technical Report”, Arxiv, 2025.

The revolution of DeepSeek

Breakthrough of Deepseek techniques

❖ **Multi-head Latent Attention (MLA)**

Attention Mechanism	KV Cache per Token (# Element)	Capability
Multi-Head Attention (MHA)	$2n_h d_h l$	Strong
Grouped-Query Attention (GQA)	$2n_g d_h l$	Moderate
Multi-Query Attention (MQA)	$2d_h l$	Weak
MLA (Ours)	$(d_c + d_h^R)l \approx \frac{9}{2} d_h l$	Stronger



[Deepseek2] Deepseek-AI. “DeepSeek-V3 Technical Report”, Arxiv, 2025.

The revolution of DeepSeek

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Deepseek-V3 (30-times Cheaper!?)

- **Auxiliary-Loss-Free Load Balancing:** Train only what's needed, save big on costs.
  - ✓ **5% of the model's parameters** were trained per token.
  - ✓ a **95% reduction in GPU usage** compared to companies like Meta.
  - ✓ **significantly lower costs,** without losing accuracy.

- **FP8 Mixed-Precision Training**
  - ✓ **Reduced memory footprint**
  - ✓ **Faster tensor operations**
  - ✓ **Less hardware requirement**

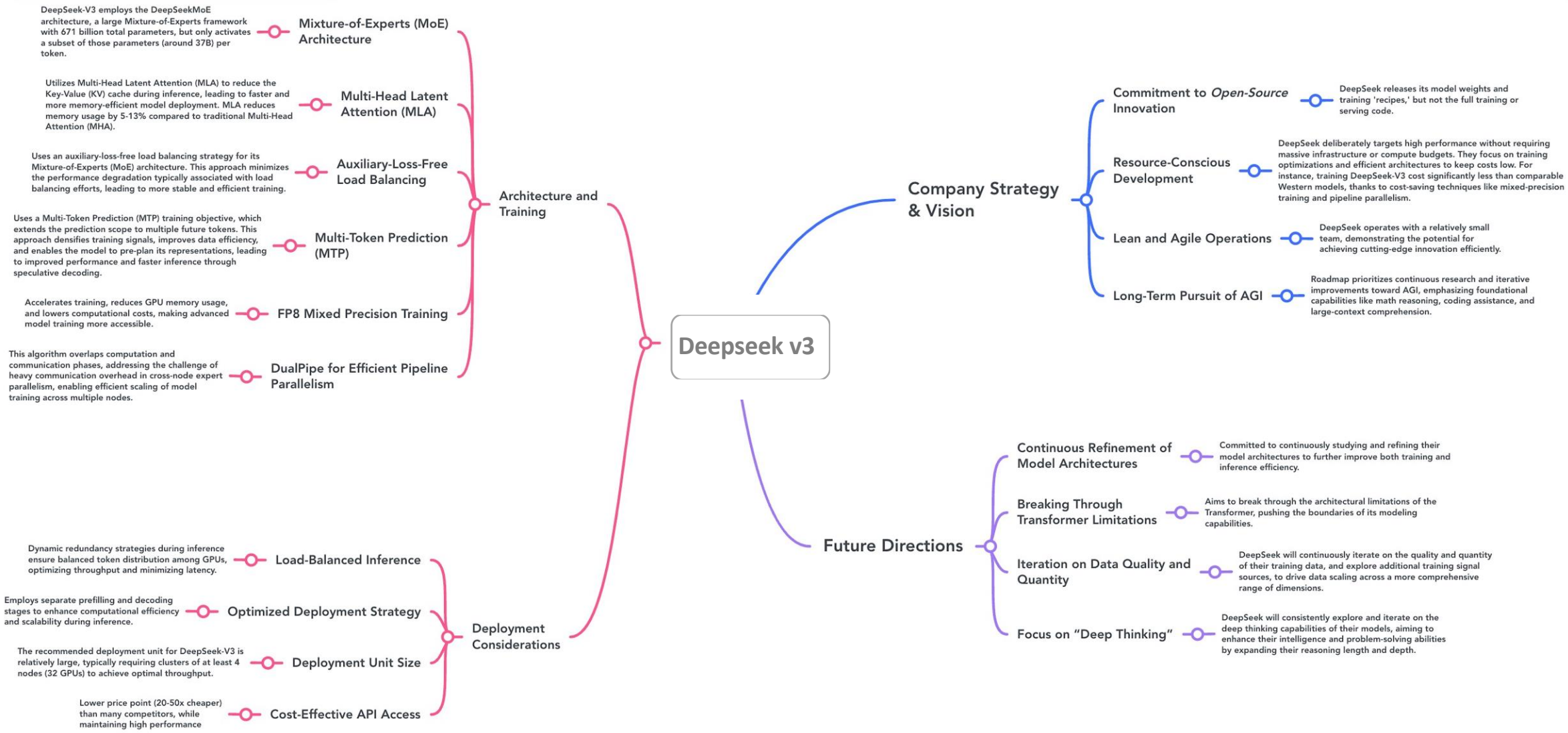
DeepSeek has priced their models 20-40x cheaper than OpenAi

Model	$/M Input Tokens	$/M Output Tokens
Deepseek-chat (V3) gpt-4o	$0.14 $2.50	$0.25 $10.00
Δ	**-94%**	**-98%**
Deepseek-Reasoner (R1) o1	$0.55 $15.00	$2.19 $60.00
Δ	**-96%**	**-96%**

Source: DeepSeek, OpenAI, Bernstein analysis

[Deepseek2] Deepseek-AI. "DeepSeek-V3 Technical Report", Arxiv, 2025.

The revolution of DeepSeek



https://gradientflow.com/deepseek-what-you-need-to-know/

From Text-Only Large Language Models To Multimodal LLMs

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Leading Enterprises in the Field of Text-to-Image Generation

Here are some of the world's top-tier enterprises in the field of text-to-image generation, along with their remarkable products.

DALLE
(from OpenAI)

Midjourney

Ideogram

Stability AI

Hunyuan AI
(from Tencent)

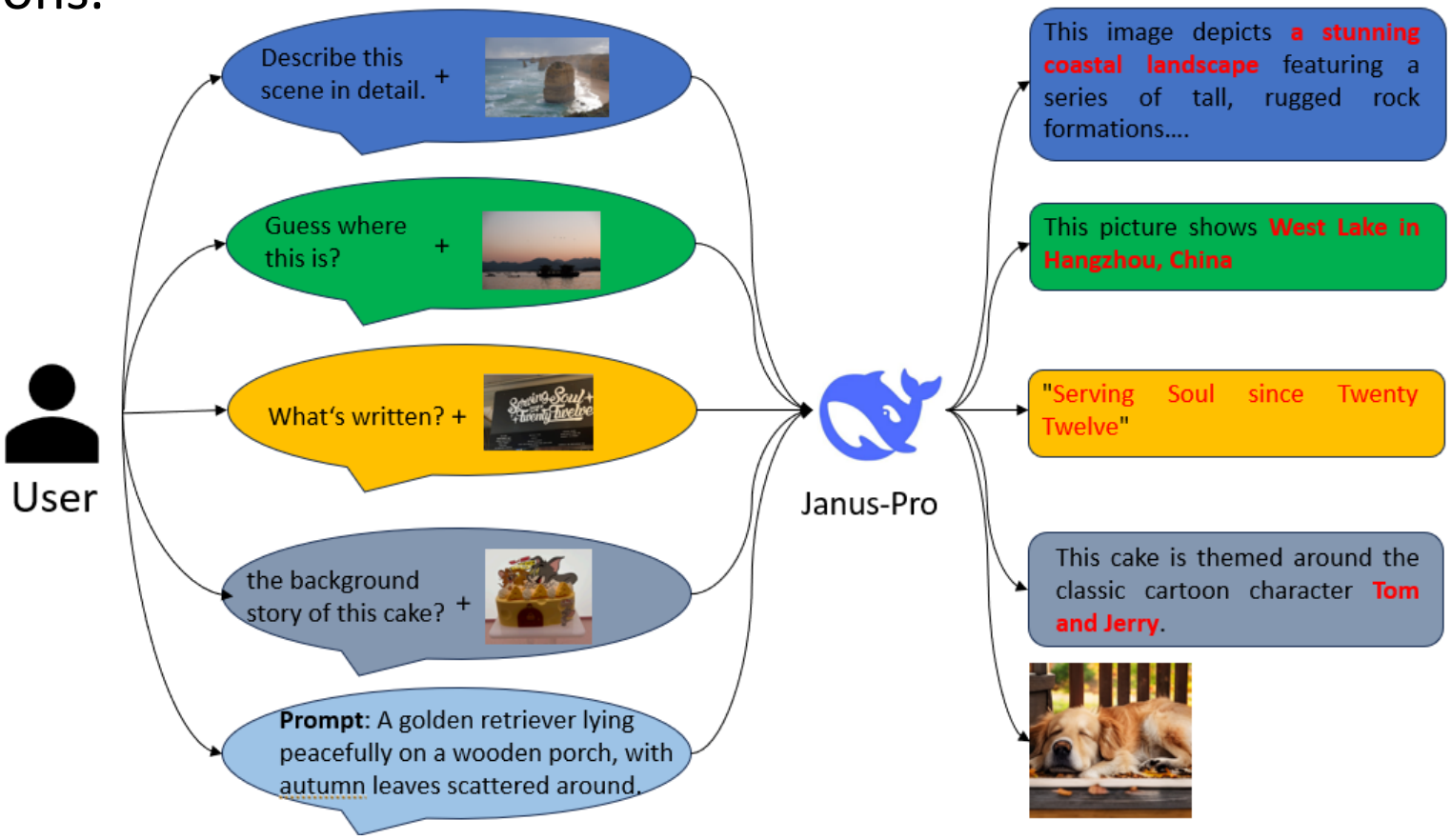
ERNIE-ViLG
(from Baidu)

Janus-Pro
(from Deepseek)

Advanced DeepSeek Multimodal AI framework: Janus-Pro

As a multimodal AI framework launched in January 2025, Janus-Pro can achieve the following functions, which endow it with strong competitiveness in the market and great potential for business applications.

- Image Description
- Landmark Recognition
- Text Recognition
- General Knowledge
- Text-To-Image Generation



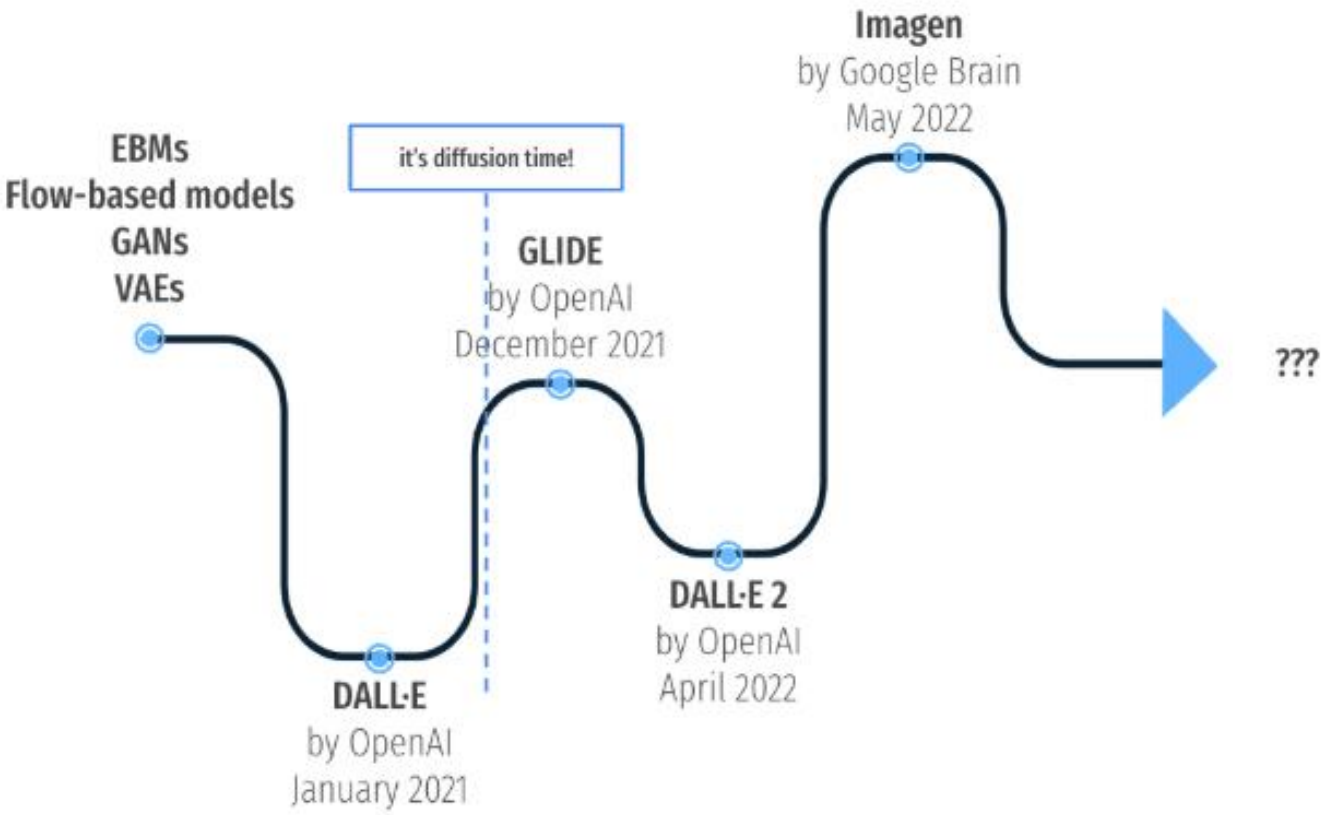
Janus Pro's diverse features

Evolution of Text-to-Image Generation Large Models

Here are a review of the technological changes in text- to-image generation before the Emergence of Janus-Pro.

Milestone Technology:

- OpenAI DALL·E: Starting from text and diving into images (2021.01)
- Diffusion: Gaussian denoising, diffusion is king (2021.12)
- GLIDE: Text-guided, directed diffusion (2022.04)
- OpenAI DALL·E 2: Complementing each other's strengths and weaknesses, a priori wonders (2022.04)
- Google Imagen: Simplify previous complexity and tipple diffusion (2022.05)



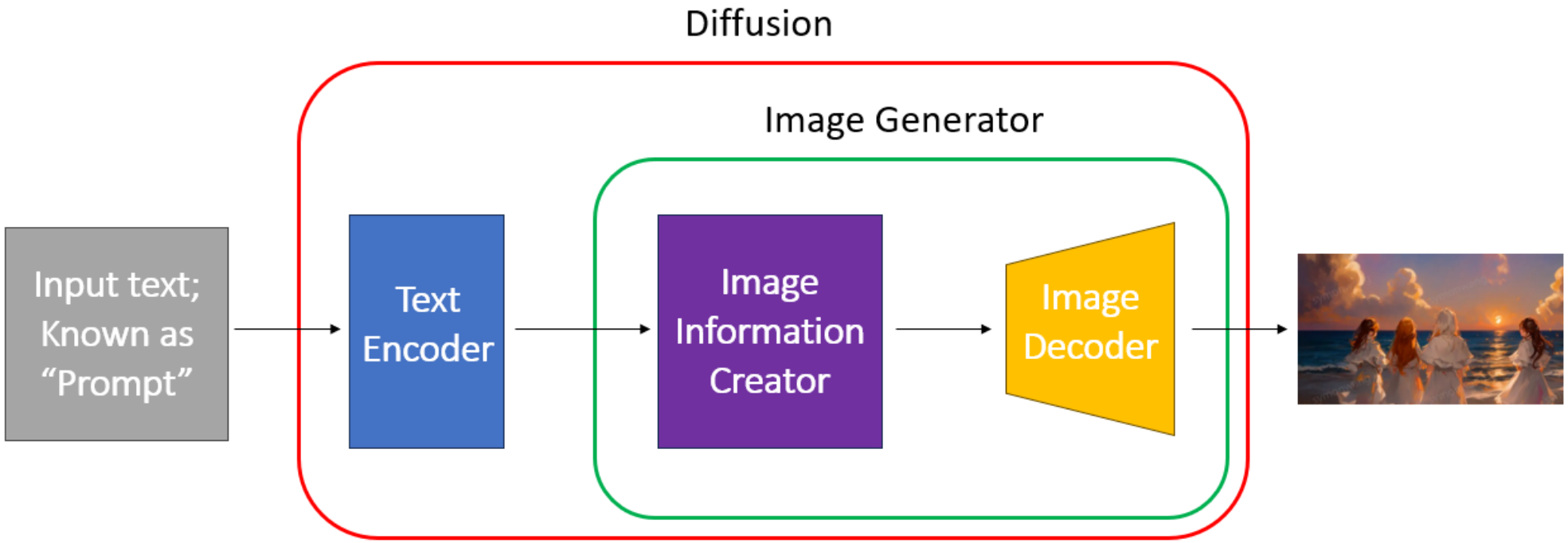
The evolution of text-to-image in recent years

"Diffusion" has become the mainstream method in text-to-image generation!

Generalized Explanation of Text-to-Image Generation Principles

Currently, the leading text-to-image generation methods largely depend on two important modules: **vision understanding** and **generation**.

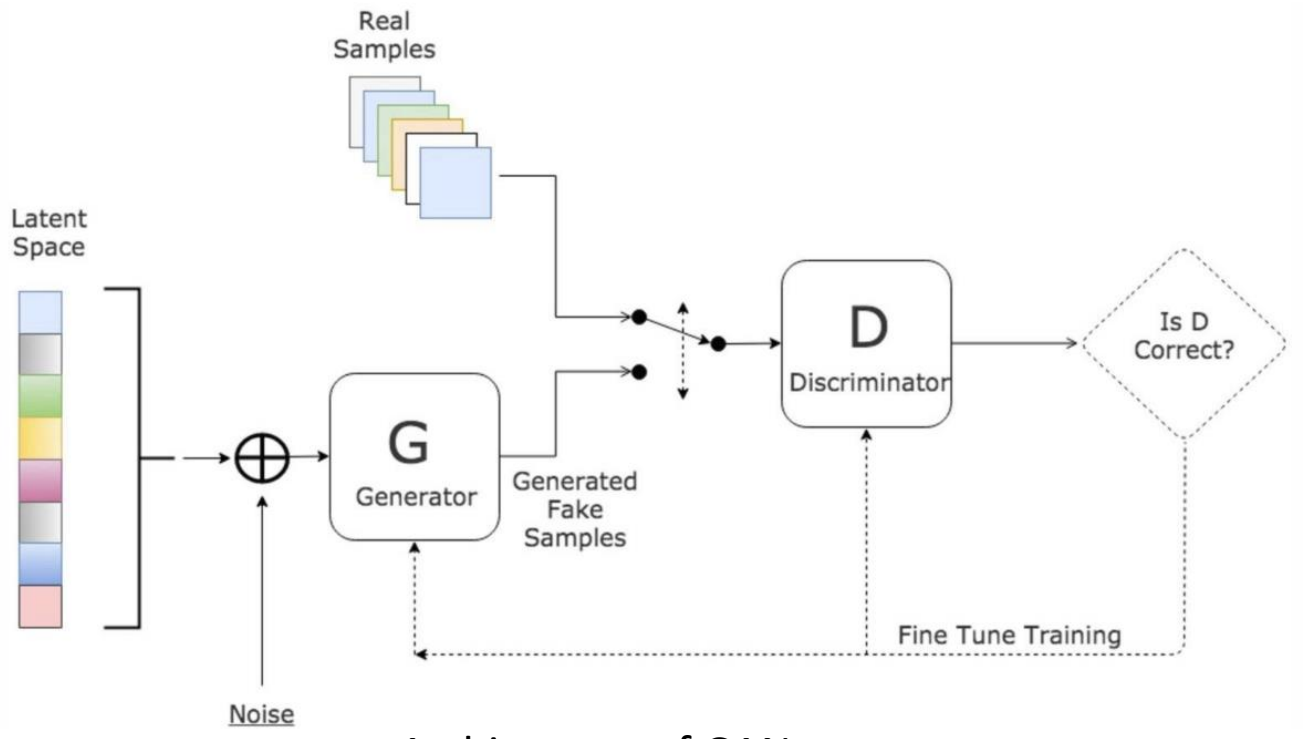
As the most popular technology text-to-image generation at present, the principle of "Diffusion" is remarkably simple and direct.



Architecture of Diffusion in text-to-image task

GAN: A General Solution Before Diffusion

Before Diffusion, Generative Adversarial Network (GAN) was the most common solution that used two neural networks, a **generator** and a **discriminator**, to compete against each other. The generator creates data, such as images or text, while the discriminator evaluates the authenticity of this data.



Architecture of GAN

Q: Why did GAN become unpopular?

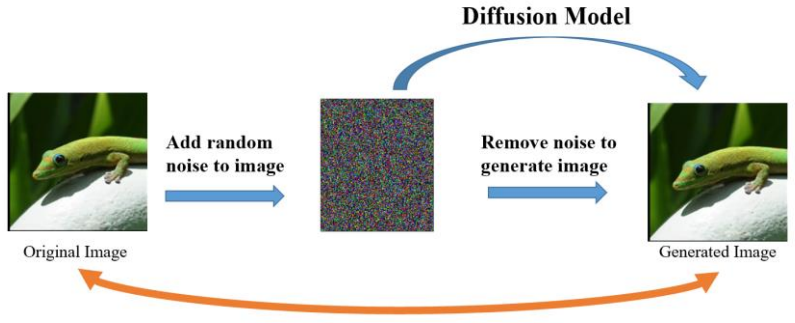
A: GAN suffers from two major reasons.

- Training instability
- Not well-suited for handling discrete data such as text.

[Goodfellow14] Ian J. Goodfellow. "Generative Adversarial Nets," NeurIPS 2014.

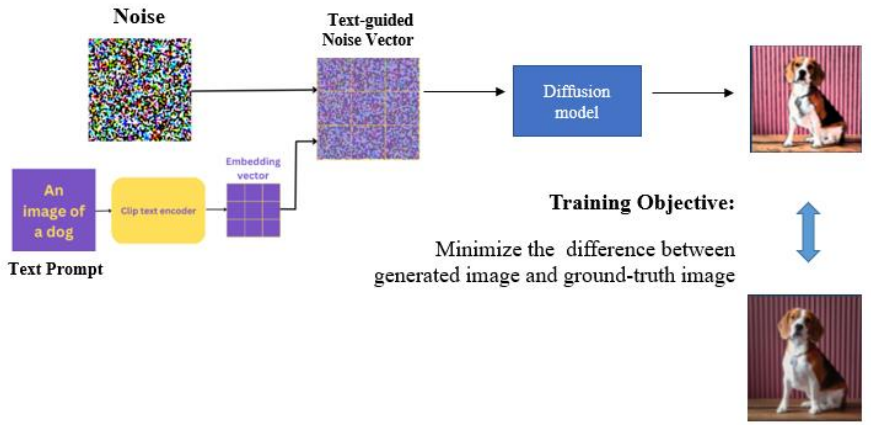
Diffusion: An important technology in Text-to-Image Task

Stable Diffusion, as the most typical model of diffusion, which was originally used to complete the task of generating images from images, converts text information into images through text encoding, image encoding, and diffusion processes (including **forward diffusion to add noise** and **backward diffusion to remove noise**).



Traditional Diffusion structure in Image-to-Image task

From Image-to-image To Text-to-image



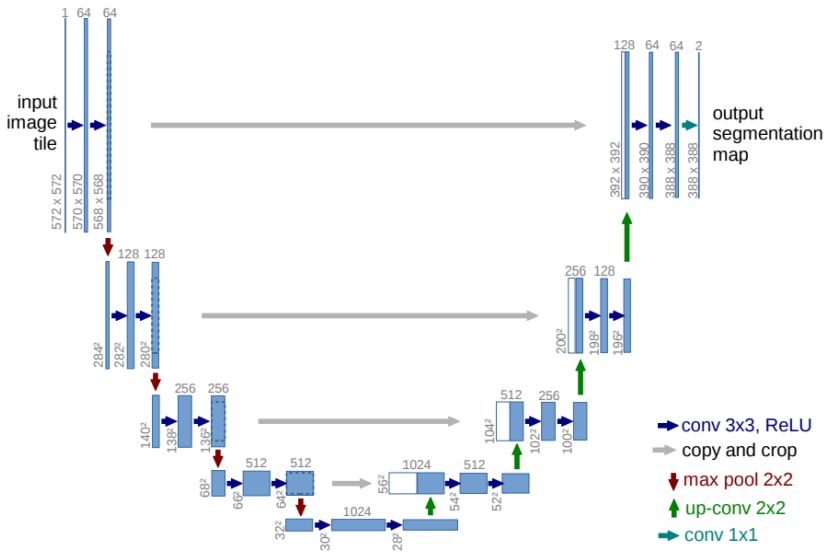
Diffusion structure in Text-to-Image task

How to deal with **noise** is the core of diffusion!

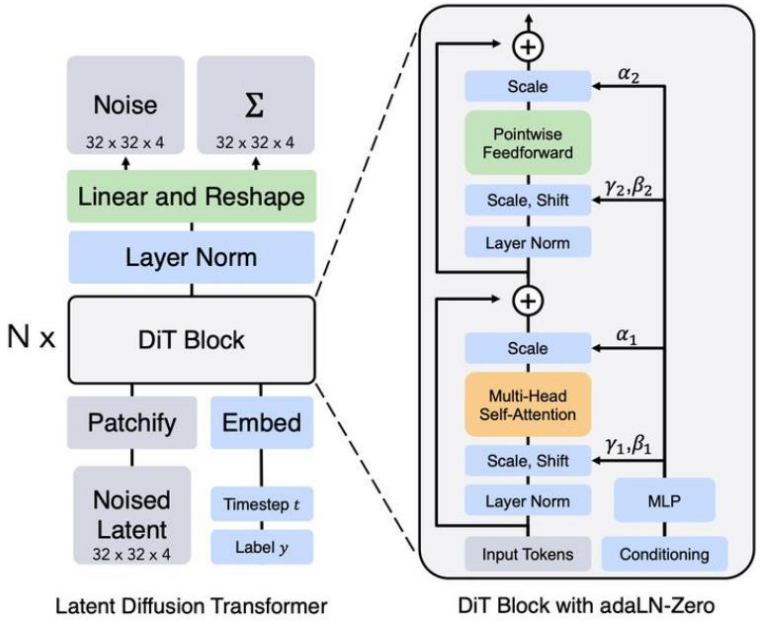
[Rombach22] Robin Rombach. “High-Resolution Image Synthesis with Latent Diffusion Models,” CVPR 2022.

Diffusion Extension: Denoising with DiT

The diffusion model is based on the experience of early work, and there is still a lot of room for improvement in network structure design. Therefore, during the denoising phase of the diffusion model, the denoising module adopted the popular transformer architecture to replace the original denoising structure (U-net), which is the **Diffusion Transformer (DiT)**.



Replaced by



Denoising structure in Stable Diffusion: U-net

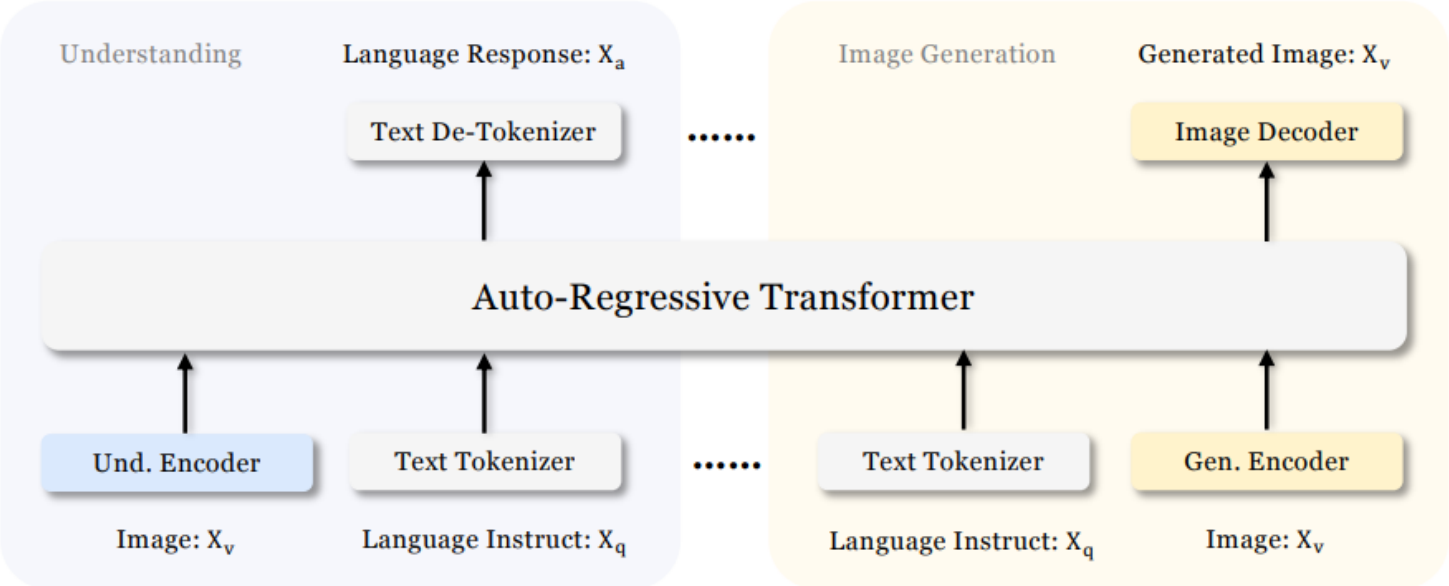
Advanced Denoising structure: DiT

[Ronneberger15] Olaf Ronneberger. “U-Net: Convolutional Networks for Biomedical Image Segmentation,” MICCAI 2015.
[Peebles15] William Peebles. “Scalable Diffusion Models with Transformers,” ICCV 2023.

Janus-Pro's Core Architecture

However, compared to most current methods that focus on how to maximize the capabilities of diffusion, Janus Pro places more emphasis on the original text-to-image generation framework: understanding and generation.

The core design principle of the overall architecture is to decouple **visual encoding** for multimodal understanding and generation.



Architecture of Janus-Pro

[Chen25] Xiaokang Chen. "Unified Multimodal Understanding and Generation with Data and Model Scaling," Arxiv, 2025.

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Thanks for your attention!